4 Evolution of Transport and Communication

Learning Outcomes

The learner:

- explains how the wheel was invented and recognizes the importance of this invention.
- identifies and describes the stages of development of transport.
- explains how improved transport led to trade and exploration of new lands.
- identifies and discusses the stages of development of communication.
- recognizes how modern technology has led to many changes in transport and communication and made the world a smaller place.

Let's Take-off

Critical Thinking

- What would our lives be like without the invention of the wheel? Talk about it class.

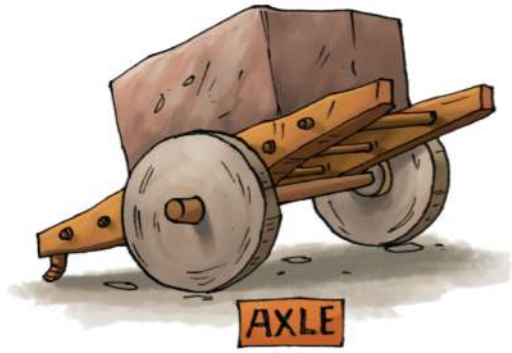
C-6.1

TP DEVELOPMENT OF TRANSPORT AND TRADE

The Wheel

The wheel was **invented** about 5,000 years ago. The earliest human beings did not have vehicles to carry large and heavy things. They probably dragged them from place to place. At some point, people realized that things that roll move faster than things that are dragged. They may have started moving heavy objects with the help of rollers made of wood. Eventually, wheels were invented. This made it much easier to move things from place to place.

Various sources tell us that the earliest vehicles to use the wheel were animal-drawn ones such as carts.



Heavy things could be moved by using logs of wood as rollers.
Wheeled vehicles that used axles made it easier to move goods.

Powering Early Vehicles

Wheeled vehicles, like carts, could transport goods for trade from place to place. The muscle power of bullocks or horses was used to move the vehicles. Goods for trade were also transported by water. Boats used the muscle power of rowers and sails that used **wind power**.

A 3000-year old stone carving from northern Iraq showing a cart

Mediterranean trading ships used by traders more than 3,000 years ago looked like this. We know what the boats looked like because boats that sank in the sea have been discovered. Talk about what you can see in the picture.

Discuss

Communication

How is the power used for vehicles these days different from the power used by early human beings for their carts and boats? C-6.1

Think and Answer

Critical Thinking

Boats with sails used wind power. What do you think would have been the disadvantages of this? C-6.1

Think and Answer

Critical Thinking

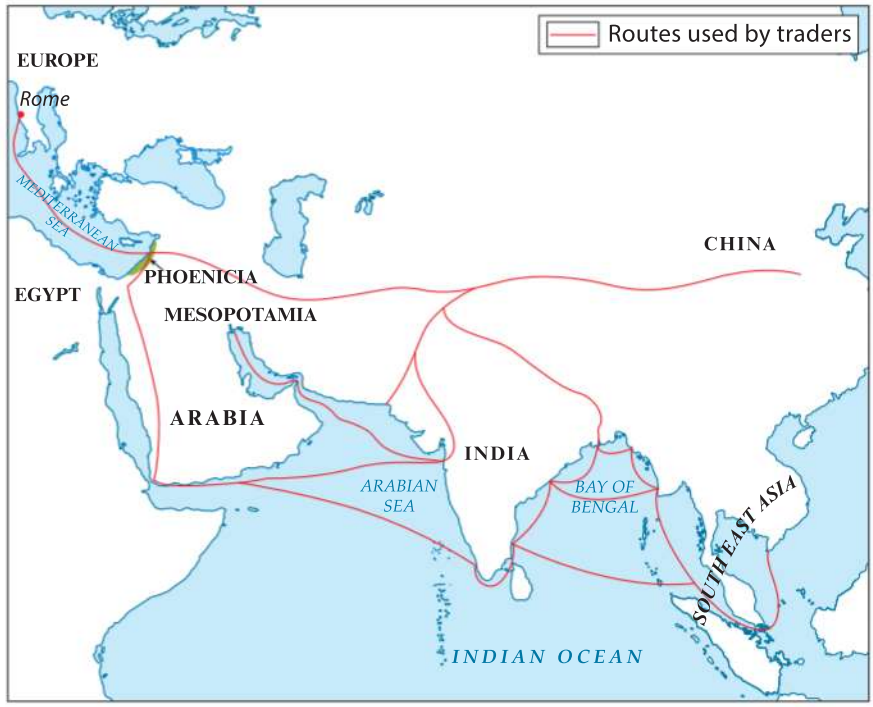
Trade journeys could be dangerous and difficult. They also took many months or years since travel was slow. Why do you think trade still continued to grow and develop? C-6.1

TP Trade with Distant Places

At first, trade was between places that were not too far away from each other. As trade grew, people began to travel to more distant places, over land and by sea. Carts could not be used for travel over land, as there were

few roads. For overland trade, goods were carried on animals like camels through the deserts of Asia and Africa. Marco Polo, a trader who lived in the 13th and 14th centuries, travelled over land from Europe to China. The journey took him about four years.

It was easier to travel to some places by sea in sailing ships. Map 1 shows some of the land and sea routes along which ancient traders travelled.



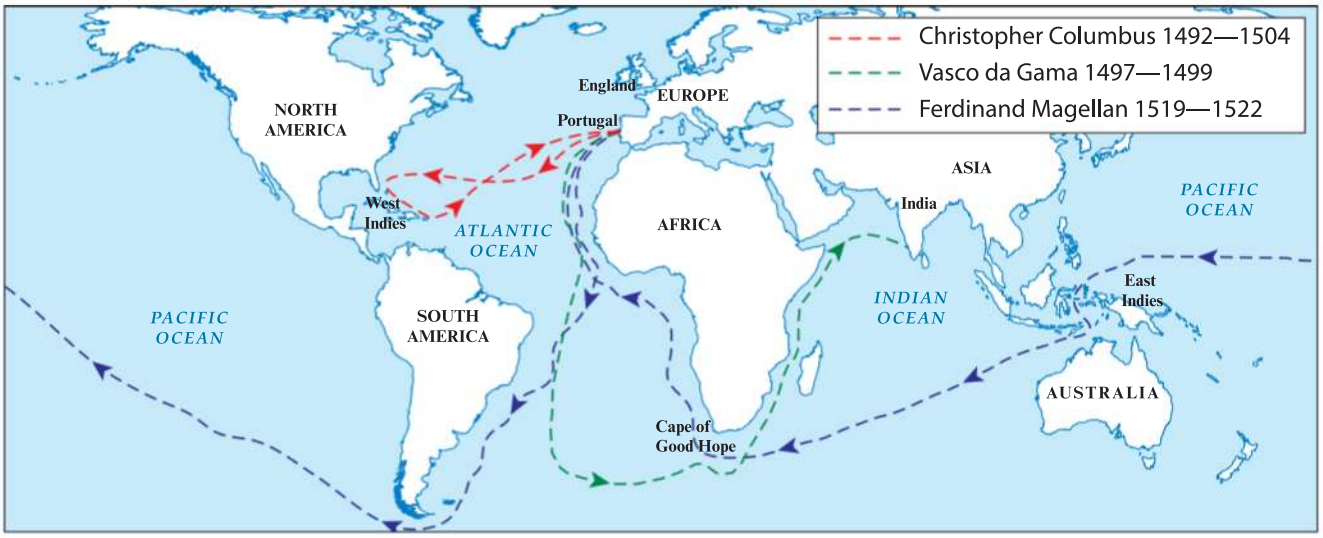
Map 1: Land and sea routes of the ancient world

Some centuries after Marco Polo, there were a number of sea journeys from Europe to different parts of the world. Map 2 on the next page shows you the routes of three famous explorers.

- **Christopher Columbus** tried to find a trade route to India by sea. He sailed west from Europe but reached the islands that we now call the West Indies.
- **Vasco da Gama** also wanted to find a sea route from Europe to India. He sailed south and travelled around Africa to reach India. His journey showed that it was possible to reach India by sea.
- **Ferdinand Magellan** sailed west from Europe, looking for a way to reach the East Indies (South-east Asia), which was also important for trade. His ships returned to Europe, having sailed around the world.

Replica of a trading ship of the 17th century—the three tall masts are for sails.

The long journeys of these explorers were possible because **shipbuilding** in Europe had improved a great deal.



Map 2: The journeys of the three explorers

Travelling Becomes Faster and Faster

For many centuries, people used muscle or wind power for vehicles. In the 18th and 19th centuries, the power of steam was discovered. This changed travel.

Knowledge of India

India was famous for its spices such as pepper, cinnamon and cloves. These were in demand in Europe for their flavours. This was one of the main reasons why the explorers were looking for a route to India.

New **steamships** used this power in engines in their ships. Sea journeys became faster. Now, a journey from England to India could be made in six weeks. Steam was also used to power **locomotives** of the first trains, which were invented in the early 19th century in England.

Steam escapes with great force from a pressure cooker.

Later in the 19th century, petroleum was discovered. Petrol and diesel are made from petroleum. Towards the end of the 19th century, the first petrol motor car was invented in Germany. In the early 20th century, the first short flight of an aeroplane using a petrol engine took place in the United States.

Did You Know?

Bertha Benz proved to her husband, Karl Benz, that the car he had invented could travel long distances. She travelled over 100 kilometres in it and proved that motor car travel was possible!

HOTS

Critical Thinking 21st Century Skills

Why did the evolution of transport take place more rapidly from the 19th century onwards? C-6.1

1400s	→	Columbus's first journey Vasco da Gama set sail from Portugal.
1800s	→	First boat to use steam Steam locomotives called Rocket First practical petrol automobile designed and built
1900s	→	First flight using petrol in USA First electric suburban train In India from Bombay (now Mumbai) First flight of an aeroplane with a jet engine
2000s	→	First commercially available car that used both electricity and petrol

A timeline showing important developments in the evolution of transport. How many 20th and 21st century methods of transport have you have seen or know about?

TP THE EVOLUTION OF COMMUNICATION

Communication means to be in touch with others through speaking, writing, reading or through other methods. We can communicate easily and quickly, even with people who may be far away, because of **technology**.

Just as travel and transport were very different in earlier times, so were the methods of communication.

Early Methods of Communication

The first writing developed in Mesopotamia over 5,000 years ago. People wrote on pieces of clay, called tablets, with 'pens' made of reeds. Paper was first invented in China several thousand years later. Even today, people use writing as an important way of communication.

For hundreds of years, people communicated with each other by writing letters. These would take a long time to reach because they were either carried by people or by carriages or boats.

About 3,000 years ago, people realized that **homing pigeons** had the ability to fly back to their homes. They could be trained to fly between two places. Pigeons were used to deliver messages quickly. Brief messages were tied to the leg of a pigeon who flew off to deliver it. Pigeons were used to send urgent messages, even during battles, right up to the 20th century.

Technological Developments in Communication

Printing press: All writing was done by hand until the 15th century CE, when the first printing press was invented in Germany. It became possible to print many copies of books and newspapers. Knowledge and information became available to many people.

The first printing press

The telegraph: In the 19th century the telegraph, a method of sending messages quickly using electricity, was developed. Important messages could be sent faster than sending a letter.

The telephone: People who were far away from each other could have a conversation. The first one took place in the 19th century.

The typewriter: The first typewriter was invented in the 19th century. Letters and documents could be typed much faster than writing them by hand.

A telephone and a typewriter of the early 20th century

Radio and television: Messages were first sent by radio in the early 20th century. News, information and entertainment could be delivered to thousands of people at the same time. Later, the first television broadcast took place.

A radio of the mid-20th century

Communication became much easier and faster than before because of these technological developments.

- The printing press made it possible to print many copies of books and newspapers, so knowledge and information became available to many people.
- The telegraph made it possible to send important messages much quicker than sending a letter.
- With the telephone, people who were far away from each other could have a conversation.
- With the typewriter, people could type letters and documents much faster than writing them by hand.
- Radio and television made it possible to deliver news, information and entertainment to thousands of people at the same time. They became very popular all over the world.

Did You Know?

The keyboard invented for the typewriter is used even today for computers, laptops and smartphones. It is called the **QWERTY** keyboard. Can you guess why?

MODERN COMMUNICATIONS

The postal system started in the 19th century, People could send letters and parcels by post. Today along with post offices, there are also courier services that deliver letters and packages.

Changes in methods of communication have happened because of developments in technology.

- **Satellites:** The first satellite was sent into space in the mid-20th century. Today communication satellites make radio, television and telephone communication possible.

- **The internet:** The world wide web of connections makes instant communication possible.
- **Mobile phones:** Mobile phones have helped many more people to be connected with others.

TP

1400s	First printing press
1800s	First electric telegraph in England
	First typewriter invented
	First telephone conversation
1900s	First long distance radio signal
	First TV transmissions
	First satellite sent into space
	First call using a large mobile phone
	Internet that we use today was created
2000s	Widespread use of smartphones

A timeline of important developments in the evolution of communication. Calculate how many centuries there are between the invention of the first printing press and the widespread use of smartphones.

A Better World

Sustainable Development Goals (SDGs)

Goal 17: Partnerships for the Goals

As modern means of transport and communication have come to be widely used, how do you think this can help governments and people to solve the problems that face us today? C-6.1

Developments in transport and communication make the world seem like a smaller place as we can travel or communicate much faster than before.

Values and Life Skills

- The wheel and technology have made transport very comfortable for us, but the fuel used in so many thousands of vehicles pollutes the atmosphere. How can we balance our need for comfortable transport with the need to have clean air?

REMEMBER

Evolution of Transport and Communication

- The invention of the wheel made transport easier.
- Travel grew because of trade; several famous journeys were made.
- Inventions in the 19th and 20th centuries made travel easier and faster.
- In ancient times, letters were carried by people, but pigeons were also used sometimes.
- Communication became easier because of several inventions in the 19th and 20th centuries.
- Modern communication is very fast because of satellites, the internet, computers, mobile phones, etc.

NEW WORDS

Locomotive	a railway engine that pulls a train
Technology	use of scientific knowledge, for example in designing new machines
Homing pigeon	a pigeon that has been trained to find its way home from a long distance away

EXERCISES

Remembering and Understanding Skills

A. Choose the correct answers.

1. The wheel was invented about 5,000/2,000/1,000 years ago.
2. The power of steam was used for aeroplanes/locomotives/cars.
3. Homing pigeons/Owls/Crows were used to send messages quickly during war.
4. The postal/telegraph/telephone system was developed to deliver letters and parcels.
5. Read the statements below and choose the correct option.

   Statement (A): Smartphones are mini computers nowadays.

   Statement (B): Smartphones are very widespread.

   (a) Both A and B are true, and B is the correct explanation of A.

   (b) Both A and B are true, but B is not the correct explanation of A.

   (c) A is true but B is false.

   (d) A is false but B is true.

B. Fill in the blanks.

1. The earliest wheeled vehicles were ..... .
2. As trade expanded in early times, people travelled to more ..... places.
3. Travel by steamships ..... the time taken for journeys.
4. In ancient times, letters sometimes took ..... to reach.
5. Communications became ..... and ..... because of several technological developments.

C. Write true or false. Correct the false statements in your notebook.

1. Things that are dragged, move as fast as things that are rolled.
2. For many centuries, the muscle power of animals was used for land transport.
3. Many inventions in communication took place in the 19th and 20th centuries.
4. Satellites do not help to send live radio, television and telephone transmissions.
5. The internet is a world wide web of connections which makes instant communication possible.

D. Match the following.

1. The travellers with their journeys of exploration.

A	B
1. Marco Polo	(a) Looked for a way to reach the East Indies.
2. Christopher Columbus	(b) Found a sea route from Europe to India.
3. Vasco da Gama	(c) Travelled overland to China.
4. Ferdinand Magellan	(d) Discovered America.

2. The centuries with the technological developments.

A	B
1. 1400s	(a) Internet that we use today was created
2. 1800s	(b) Widespread use of smartphones
3. 1900s	(c) First typewriter invented
4. 2000s	(d) First telephone conversation
5. 1800s	(e) First call using a large mobile phone
6. 1900s	(f) First printing press

E. Answer the following questions.

1. How were goods for trade transported in ancient times?
2. What power was used by early ships and how was this different from the power used in the ships of the 19th century?
3. How has travel between India and Britain changed in the type of transport and the time the journey takes? Explain in two or three sentences.
4. What does communication mean?
5. Arrange these 'power' words in the right time order: electricity, muscle, petrol, steam
6. Write one sentence each about the advantages of the following inventions—printing press, telephone and radio.

DO AND LEARN

TP

A. Picture Study

21st CENTURY SKILLS

This is a page from a book written and decorated by hand. Look at it carefully and answer the questions:

1. How long would it have taken to write such books?
2. Would such books have been easily available to most people?
3. Why was the printing press an important invention?

C-6.1 Communication SI History

B. On Your Own

Map activity

On an outline map of the world, mark the journeys of the three explorers , Christopher Columbus, Vasco Da Gama and Ferdinand Magellan.

C-5.3 Map marking

TP

C. In a Group

Work together to arrange these events and their dates in the right order:

Transport		Communication	
1885	Karl Benz's petrol automobile	1925	First live TV transmission
1903	First flight at Kitty Hawk	1440	First printing press
1271	Marco Polo starts journey	1876	First telephone conversation
1807	First boat to use steam	1901	First long distance radio signal
1492	Columbus's first journey	1868	First typewriter
1519	Magellan starts journey	1957	First satellite sent into space
1497	Vasco da Gama starts journey	1837	First electric telegraph in England
1829	Steam locomotive called *Rocket*		

D. Pair and Share: Imagine you and your partner have the first-ever conversation over a telephone. What would you have said? Discuss and enact it.

C-6.1 Communication

Weblinks For more information, go to:

- **Bertha Benz: The First Driver:** https://www.youtube.com/watch?v=JBL_G-C51Dk (Accessed on 31 July 2025)
- **The Invention of the Printing Press:** https://www.youtube.com/watch?v=BhbI0eDGmsA The Invention of the Printing Press (Accessed on 31 July 2025)
- **The Fascinating History of Ice Cream:** https://www.parsidairyfarm.com/blogs/news/history-ice-cream-evolution?srsltid=AfmBOoqvhWOAcL1rQN9nhJGKRF3Xxtn7pTHXbsrHkRe25b-TVRydFz1A (Accessed on 31 July 2025)

What I Learnt

Self-review

Colour the smileys to show how well you have understood the concepts.	Completely	Mostly	Need help
• I can explain how the wheel was invented and recognize the importance of this invention.	☺	☺	☺
• I can identify and describe the stages of development of transport.	☺	☺	☺
• I can explain how improved transport led to trade and exploration of new lands.	☺	☺	☺
• I can identify and discuss the stages of development of communication.	☺	☺	☺
• I can recognize how modern technology has led to many changes in transport and communication and made the world a smaller place.	☺	☺	☺